

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and deployed **ML algorithms** and **AI applied research solutions** using **Python (scikit-learn, pandas, NumPy)** and **transformer neural networks** to measure real customer impact; built end-to-end **classic ML and deep learning models** for production environments, achieving **92% accuracy** on classification tasks and reducing inference latency by **40%**.
- Designed and implemented **data science pipelines** using **Python (Flask, FastAPI)** and **SQL (PostgreSQL, MySQL, Oracle)** to support **AI applied research and innovation**; engineered **ETL workflows** for feature engineering and model training, processing **500K+ records** monthly with **99.2% data accuracy**.
- Deployed **ML models** to production using **AWS (EC2, S3, Lambda, CloudWatch)** and **Docker containerisation**; implemented **CI/CD pipelines** via **Jenkins** to automate model validation and release; established **monitoring dashboards** using **CloudWatch** to track real-time model performance and business impact metrics.
- Created interactive **AI dashboards and data visualisation** using **Power BI (DAX, Power Query)** and **Tableau** to communicate **ML model results** and **algorithm performance** to business stakeholders and technical teams; built **data modelling architectures** supporting real-time **AI infrastructure monitoring** and decision-support reporting.
- Trained and supported the AI ecosystem through **training materials, advisory services**, and technical documentation; collaborated cross-functionally with industry and public-sector partners to adopt **AI applied research methodologies**; conducted **statistical analysis** and **hypothesis testing** to validate model efficacy and guide algorithm selection.
- Managed **AI infrastructure** and system reliability through **production support and incident response** for deployed **ML models and neural networks**; implemented **SLA management** and alerting via **Splunk and Dynatrace**; optimised algorithm performance and **model evaluation metrics** (precision, recall, F1-score) to improve real customer outcomes by **35%**.

SKILLS

Machine Learning & AI – ML algorithms, classic machine learning, transformer neural networks, supervised learning, unsupervised learning, deep learning, model evaluation metrics, algorithm optimisation.

Data Science & Analytics – Python (scikit-learn, pandas, NumPy), statistical analysis, hypothesis testing, exploratory data analysis, R/RStudio, A/B testing, time-series analysis, regression analysis.

Data Engineering & Pipelines – ETL workflows, data pipeline development, feature engineering, data modelling, data warehousing, SQL (PostgreSQL, MySQL, Oracle), Python (Flask, FastAPI).

Business Intelligence & Visualisation – Power BI, Tableau, DAX, Power Query, interactive dashboards, data visualisation, business reporting.

AI Infrastructure & Production – AWS (EC2, S3, Lambda, CloudWatch), Docker containerisation, CI/CD (Jenkins), Git, production support, SLA management, incident response, monitoring (CloudWatch, Splunk, Dynatrace).

Soft Skills & Domain – training and advisory delivery, cross-functional collaboration, requirements gathering, stakeholder communication, applied AI research, European Digital Innovation Hub standards.

Certifications & Qualifications – MSc Data Analytics (NCI, Dublin), Full Stack Java Spring Boot & Angular, Web Development (UC Davis), Python Programming (University of Michigan), IELTS Academic Band 7.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin	02/2026
BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India	01/2021

PROJECTS

AI-Driven Customer Impact Measurement & Model Deployment Platform (HCL Technologies / CeADAR Applied Research)

- Engineered end-to-end **ML model development pipeline** combining **classic ML algorithms and transformer neural networks** using **scikit-learn, TensorFlow/PyTorch, and Python data stack (pandas, NumPy)**; built supervised and unsupervised models for industry customer use-cases, achieving **94% precision and 89% recall** on validation datasets.
- Developed **real-time model monitoring and evaluation infrastructure** using **AWS (Lambda, CloudWatch, S3)** for deployed **ML models**; implemented automated **algorithm performance tracking** dashboards in **Power BI and Tableau** measuring accuracy, latency, and business impact; reduced model inference time by **45%** through Docker containerisation and Lambda optimisation.
- Deployed **AI applied research solutions** to production via **CI/CD pipelines (Jenkins, Git)**; designed **data pipelines and ETL workflows** using **PostgreSQL and Python (FastAPI)** to ingest customer transaction data, engineer features, and feed model training—processing **2M+ monthly records** with **99.5% data quality assurance**.
- Measured and demonstrated **real customer impact** through **statistical analysis and hypothesis testing**; conducted A/B testing and time-series analysis to validate algorithm effectiveness; documented results in **training materials and advisory reports** for ecosystem partners, enabling **3+ industry clients** to successfully adopt **AI applied research methodologies**.
- Managed **AI infrastructure reliability** through **production support, incident response, and SLA management**; implemented comprehensive monitoring using **CloudWatch and Splunk**; achieved **99.8% model uptime** and **15-minute P1 incident response** for critical deployed algorithms.

SEPA Payments AI Risk Scoring & ISO 20022 Compliance Analytics (HCL Technologies Banking Domain)

- Built **ML classification models** using **scikit-learn and statistical methods** to predict high-risk SEPA payment transactions; integrated with **SWIFT MT/MX message flows and ISO 20022 message structures**, improving fraud detection accuracy by **38%** while maintaining **99.2% transaction throughput** for cross-border payments.
- Developed **Power BI dashboards** with **DAX measures and Power Query ETL** to visualise **ISO 20022 compliance metrics and payments reconciliation workflows**; enabled regulatory audit teams and business stakeholders to track **GDPR data protection compliance** and tokenization processes in real-time.
- Engineered **data pipelines** using **Python (pandas, NumPy) and SQL (PostgreSQL, Oracle)** to extract, transform, and load **SEPA/instant payments transaction data and Open Banking / PSD2 message logs**; supported **change-control management** and UAT coordination for regulatory implementation across interconnected banking systems.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)